Thakaa at AlexandriaX-2026: Who Gets the Final Say? Assigning Output Authority in Dialectal Arabic Machine Translation and Error Detection

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Abstract

We report first-place systems for all four evaluated AlexandriaX-2026 settings: both English-to-dialect translation tracks, cross-dialect translation, and translation-error detection. In unconstrained translation, our minimum Bayes risk (MBR) selection restricts output authority: four local systems can supply the translation, while seven additional systems contribute only reference evidence. Candidate-constrained MBR beats the strongest complete candidate in all 13 dialects and gains 1.2554 macro spBLEU overall, reaching 33.4891. It also surpasses a retrospective oracle selecting the best whole candidate system per dialect. In the constrained track, country-routed LoRA adapters over a shared NileChat-3B-Base backbone incorporate the released Libyan and Sudanese supervision absent from the training and development splits, raising macro spBLEU from 29.5969 to 30.4233. Post-evaluation, a single adapter trained on all released Alexandria data reaches 30.4733 macro spBLEU; averaging model weights across three training seeds raises this to 30.9726 while retaining one model for inference. For cross-dialect translation, nonparallel target-dialect retrieval guides deliberation, raising Gemma-4-31B-IT without fine-tuning from 24.6898 to 28.4011 spBLEU. For translation-error detection, GPT-5.5 proposes semantic error regions, while MARBERTv2 controls their character boundaries and corroborates additional spans, raising Overall from 0.4691 to 0.4982.

1 Introduction

Dialectal Arabic systems must preserve meaning while realizing a specified target variety, despite uneven supervision across dialects; translation-error detection additionally requires exact character-level localization. AlexandriaX-2026 evaluates these challenges through English-to-dialect conversational translation, cross-dialect translation, and character-level detection and classification of MT errors (El Mekki et al., 2026a). We participated in every task and both Subtask 1 tracks, placing first in all four evaluated settings (Table 1).

The resulting systems combine fine-tuned generators, retrieved examples, selectors, and span controllers in different ways. We use *output authority* to distinguish components allowed to emit or finalize the scored prediction from those that only condition, propose, or score it. In unconstrained translation, extra systems help select an answer without being allowed to supply it (Figure 1).

Controlled comparisons yield several findings. In unconstrained translation, the reference-only jury improves candidate-constrained MBR, whereas making the same systems selectable forfeits part of that gain. For NileChat-3B-Base, beam search produces the largest measured inference-time improvement, and authentic Libyan and Sudanese examples outperform an equal-size duplicate-row control. A separate subdialect-conditioning comparison shows that derived labels help chiefly where country-level labels collapse several annotated varieties. In cross-dialect translation, nonparallel target-dialect retrieval improves spBLEU, with a controlled gain from 16 to 32 examples.

Task	Setting	Primary	chrF++	Rank
ST1	Constrained	30.4233	45.4250	**1/6**
ST1	Unconstrained	33.4891	48.1320	**1/6**
ST2	Cross-dialect	28.4011	42.1509	**1/4**
ST3	Error detection	0.4982	n/a	**1/6**

Table 1: Official blind-test results. ST: subtask; rank: position among participating teams. Primary: spBLEU for ST1/ST2 (SacreBLEU, FLORES-200 tokenizer; Post, 2018; NLLB Team et al., 2022); Overall for ST3, averaging overlap F1 0.5426 and class F1 0.4538. ST1 averages 13 dialects; chrF++ (Popović, 2017) is secondary.

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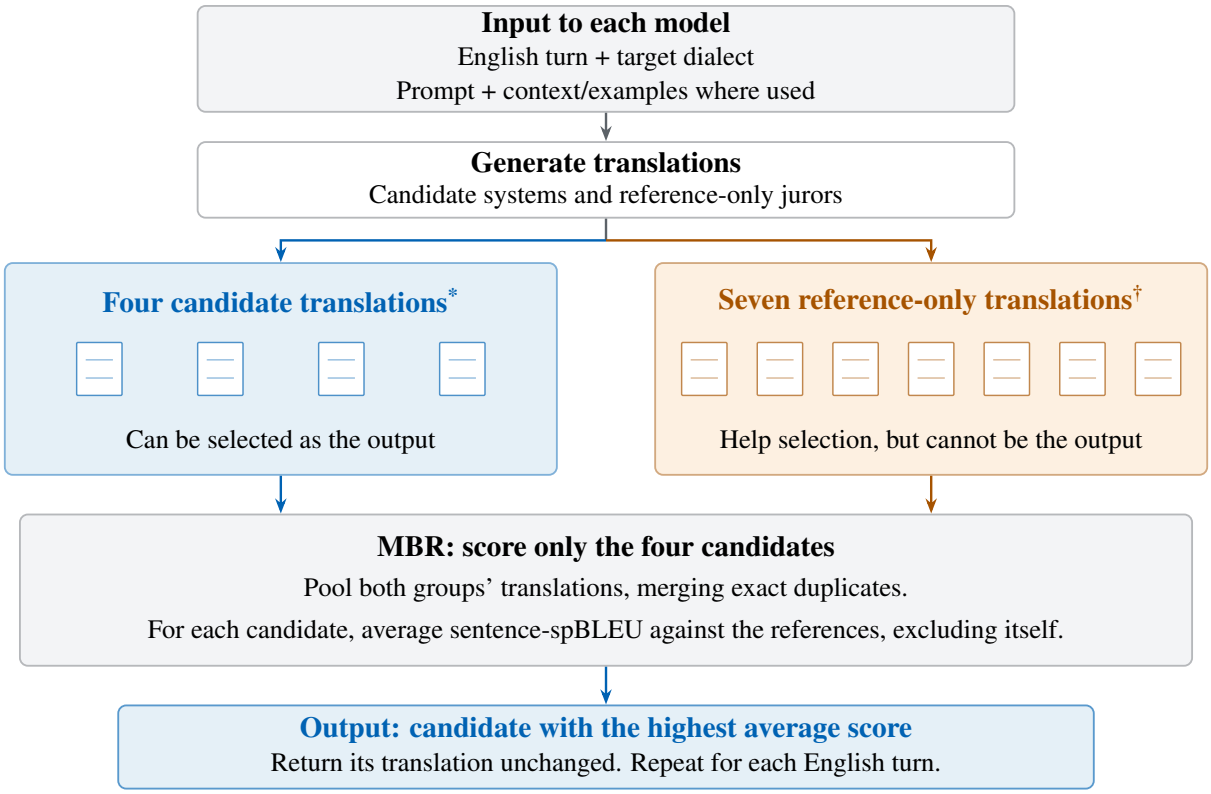


Figure 1: Candidate-constrained MBR: additional translations help choose the answer, but only a candidate can supply it. For each turn, we select the candidate with the highest average sentence-spBLEU against the other unique translations (Eq. 1). *Candidate models and training recipes: Table 2. †Reference-only jury models and prompts: Appendix A. The submitted system reaches 33.4891 macro spBLEU, 1.2554 above the best single candidate system and 0.8145 above choosing the best whole system separately for each dialect (32.6746). This retrospective comparator uses blind-test scores to choose one system for all turns in each dialect; MBR selects separately for each turn without hidden references (Section 2.2).

In translation-error detection, missed spans and imprecise boundaries dominate the remaining errors, rather than category selection once a match exists. Consistent with this diagnosis, the blind-test gains come from encoder-guided boundary widening and agreement-gated span additions while retaining GPT-5.5 as the semantic proposer.

2 Subtask 1: English-to-Dialect Translation

Subtask 1 translates each English conversational turn into one of 13 country-specified Arabic varieties. Inputs include conversation metadata and preceding dialogue context; the 14,459-turn blind test is ranked by macro spBLEU across countries (El Mekki et al., 2026a,b). We address complementary candidate selection in the unconstrained track and constrained coverage for Libyan (LY) and Sudanese (SD) Arabic, which are absent from the official training and development splits.

2.1 Shared conditioning: metadata-derived subdialect labels

The Alexandria dataset study reports stable within-country subdialect difficulty across models but mixed gains from general metadata (El Mekki et al., 2026b). We isolate the target label itself: training and development provide fine-grained annotations, whereas blind-test inputs generally identify only the country. In training, identifier prefixes map deterministically to the subdialect in all 129 observed country-prefix pairs. The mapping labels 98% of blind-test conversations in the eleven represented countries, uses a country-level fallback elsewhere, and changes the country label in 24% of prompts. All submitted Subtask 1 systems use this benchmark-specific label.

Between two near-matched official submissions of our strongest 12B candidate (same model and greedy decoding; different batch sizes), switching the prompt from country to subdialect raises

Candidate system	Adaptation, data, and decoding	spBLEU
Gemma-4-12B-IT CPO	rank-16 QLoRA SFT: train+dev+public test + AraBench (2,500 LY + 2,000 SD examples); CPO: 13,049 chrF++-ranked gold-vs-best pairs; terminal-punctuation normalization	32.0318
12B CPO + ICL	same trained 12B weights + k=8 Alexandria training-split examples where available (none for LY/SD), retrieved by identifier group (BM25 fallback)	32.2235
Gemma-4-E4B-IT SFT	rank-32 LoRA SFT: train + public test + AraBench LY/SD (no dev); beam-8	32.2336
NileChat-3B-Base soup	3× rank-32 LoRA (train+dev), uniform weight average; beam-8, MR/MA training-split demonstrations	29.5969

Table 2: The four systems producing the selectable translations in Figure 1, with their adaptation recipes and official standalone blind-test spBLEU (Gemma Team, 2026; Xu et al., 2024; Sajjad et al., 2020; El Mekki et al., 2025). SFT: supervised fine-tuning; CPO: contrastive preference optimization; ICL: in-context learning. All use metadata-derived subdialect labels. The NileChat row is the original train+development soup, not the routed constrained submission or the post-evaluation full-data soup.

macro spBLEU from 30.61 to 31.79 (+1.18): the five multi-subdialect countries supply +1.14 (OM +7.47 to SA +1.01) and the other eight only +0.04, suggesting that batch-size drift is small. In an Oman-only development ablation, the subdialect-trained adapter improves only when the matching fine-grained label is also supplied at inference (Figure 2(a)). These comparisons support preserving target-label granularity when labels are available. The identifier-prefix mapping is benchmark-specific; these experiments evaluate supplied or metadata-derived labels, not subdialect identification from text.

2.2 Unconstrained system: candidate-constrained MBR

The candidates follow separate improvement paths (Table 2; Appendix A): contrastive preference optimization (CPO) on a fine-tuned Gemma-4-12B-IT; identifier-matched demonstrations for its in-context-learning (ICL) variant (Agrawal et al., 2023); fold-in supervised fine-tuning (SFT), which adds released public-test rows, and beam search for the compact E4B (effective 4B); and the constrained-track NileChat soup. Their outputs are complementary: post-evaluation demotion of any candidate costs 0.1332 to 0.4175 macro spBLEU, and the soup wins 12.6% of turns despite its lowest standalone score, so all four supply useful output hypotheses (Appendix C.4).

For each turn, the candidates form the selectable set $\mathcal{C}(x)$; a fine-tuned Gemma-4-E2B-IT system (effective 2B) and six prompted GPT/Gemini systems form a reference-only jury $\mathcal{J}(x)$. Exact duplicates across all eleven outputs collapse into single hypotheses, giving one pool $\mathcal{H}(x)$; with $\mathcal{H}_y = \mathcal{H}(x) \setminus \{y\}$, we select

$$\hat{y} = \arg\max_{y \in \mathcal{C}(x)} \frac{1}{|\mathcal{H}_y|} \sum_{r \in \mathcal{H}_y} \text{spBLEU}(y, r). \quad (1)$$

Each distinct juror hypothesis can affect agreement, while only a local candidate can be returned. The rule follows MBR decoding and system combination (Kumar and Byrne, 2004; Eikema and Aziz, 2022; Freitag et al., 2022; González-Rubio and Casacuberta, 2013; Farinhas et al., 2023). Unlike recombination or preference optimization (Jon and Bojar, 2023; Wang et al., 2026), it selects among fixed outputs. Building on multi-prompt and dialectal-Arabic MBR (Heineman et al., 2024; Alali and Issam, 2026), we evaluate restricting outputs to the four local candidates.

Selection versus additional reference evidence. With the four candidate outputs fixed, a post-evaluation control using only those candidates as mutual references scores 33.0925, exceeding the strongest standalone candidate, E4B at 32.2336, by 0.8589. Adding the seven reference-only jurors raises this to the submitted 33.4891, a further 0.3966. Of that jury increment, E2B adds 0.1025 before the API jurors are introduced; the six API jurors then add 0.2940 and improve 10 of 13 countries. Thus, most of the 1.2554-point gain is already obtained by selection within the local pool, with additional reference evidence improving that selection. This is a conditional score decomposition for fixed candidates and utility, not an independent attribution to each model: the jury-family factorial shows a positive interaction (Appendix C.4). Appendix F consolidates the main comparisons across all four settings, identifying the reference configurations, score changes, and evaluation phases.

The selector also scores 0.8145 above a retrospective dialect-level system oracle that selects one complete candidate per dialect using blind-test scores (32.6746); its margin is positive in all 13 dialects, demonstrating within-dialect complementarity (Table 13). In a post-evaluation control, granting the jurors output eligibility over the same hypothesis pool and utility scores 33.2998, forfeiting 0.1893 of the gain (Figure 3).

In a separate post-evaluation experiment, the MBR system with the full-data soup (Section 2.3) replacing the original NileChat candidate reaches 33.5246 macro spBLEU, slightly above the submitted system's official 33.4891 (Appendix C.4).

2.3 Constrained system: shared-backbone LoRA routing

The constrained track permits only Alexandria data and caps the system at 5B raw parameters. Track policy admits all released data; the public-test split is the only released LY/SD supervision. Using the released country field, we route LoRA adapters (Hu et al., 2022) over one shared 3.086B `NileChat-3B-Base`: the default is the uniform three-seed soup of Table 2 (Wortsman et al., 2022), while LY/SD use one same-recipe adapter trained on train plus public test.

Both routes decode with beam-8 in bf16, condition on the derived subdialect label (Section 2.1), and reuse preceding Arabic outputs as self-context. Mauritanian (MR) and Moroccan (MA) inputs also receive three same-dialect training-split exemplars in the training template (+0.85/+0.81 development spBLEU; EG, TN, and YE controls flat or worse). The backbone and four deltas total approximately 3.33B parameters. Routing changes only the 2,224 LY/SD turns, leaving the other eleven countries byte-identical, so the gain from 29.5969 to 30.4233 comes entirely from LY (21.85 to 26.72) and SD (22.29 to 28.17) (Table 4; Figure 2(b)).

A matched post-evaluation ablation isolates the value of authentic LY/SD supervision (Appendix C.1). Its baseline is a single seed-42 adapter trained on training and development data, not the three-seed soup used in the submission. Adding public-test LY/SD raises macro spBLEU from 29.2323 to 30.1179, while an equal-size duplicate-row control falls to 29.0971. Training on all released Alexandria data reaches 30.4733, versus 28.8483 for full-budget duplication, supporting the value of new supervision rather than additional optimizer updates alone.

Weight averaging after full-data training. A further post-evaluation comparison repeats the full-data recipe with seeds 42, 123, and 7. Individual models score 30.4426 to 30.4733; averaging their merged weights yields 30.9726, a gain of 0.4993 over the best seed, and improves spBLEU over the strongest individual seed in every dialect (Table 6). This soup is decoded as one 3.086B-parameter model, without routing or an inference-time ensemble. It is distinct from the submitted train+development soup. Together, full-data training and weight averaging give a stronger single-model recipe than the routed submission, although the different training configurations do not isolate the effect of removing routing. Further beam-search and conditioning controls appear in Appendix C.1.

3 Subtask 2: Cross-Dialect Translation

Subtask 2 comprises 3,574 retail-banking translations across all 30 directed pairs among six Arabic dialects (El Mekki et al., 2026a; Malaysha et al., 2024). No task-specific training or development references are released, leaving target terminology, morphology, and spelling underdetermined even when the banking intent is preserved.

3.1 System: target-dialect retrieval and guided generation

For a query q from dialect A to B, the retrieval bank contains released blind inputs written in B, treated as unpaired target-dialect usage examples with no cross-dialect alignment used in retrieval. We embed unique sources with multilingual E5-Large (Wang et al., 2024). To suppress likely cross-dialect counterparts, we discard candidate c when q is among its 32 nearest dialect-A sources, then rank survivors by a bitext-mining relative margin (Artetxe and Schwenk, 2019; Equation 2). The top 32 enter the prompt as usage evidence; none is identified or presented as a translation of q (Tan et al., 2024).

A fixed prompt asks a frozen Gemma-4-31B-IT, decoding greedily, to weigh source-faithful, target-idiomatic, and retrieval-aligned renderings into one synthesis, a single-call latent variant of reasoning prompts for MT (Nguyen and Xu, 2025). The pipeline never queries hidden reference data or intent labels. An exact-copy guard regenerates whenever the normalized output copies an active example, falling back to the clean no-retrieval out-

put on persistent copies (Appendix D.1).

3.2 Results and integrity analysis

The retrieved messages expose target-dialect vocabulary, morphology, spelling, and style. Consistent with the model exploiting this evidence, official spBLEU rises from 24.6898 without retrieval to 27.9496 at k=32; the 16-to-32 comparison holds copy handling fixed (Appendix D.2). Guided generation brings the final system to 28.4011 (+0.45; Table 14).

The method is transductive over the released blind sources. A retrospective audit against public ArBanking77 counterparts (Jarrar et al., 2023) found at least two recoverable query counterparts among 3,574 final prompts, including one case of wording reuse. We exclude overlapping ArBanking77 rows from independent evaluation claims; Appendix D.3 documents the audit and exploratory analyses.

4 Subtask 3: Error Span Detection and Classification

Subtask 3 requires translation-error localization and six-way classification under Linguistically Motivated Multidimensional Quality Metrics (LQM) (Magdy et al., 2026) over 1,125 training, 138 development, and 145 blind-test examples in five directions (El Mekki et al., 2026a). Overall averages character-overlap F1 and pooled error-class F1, so systems must both localize errors and label matched spans; the task sits between LLM error analysis (Kocmi and Federmann, 2023) and span-supervised evaluation (Guerreiro et al., 2024).

4.1 System: semantic proposal and span control

Given the English source and its Arabic translation, GPT-5.5 proposes semantic error regions by reproducing the translation with inline category tags, which we map deterministically to character offsets. A 162M-parameter MARBERTv2 encoder (Abdul-Mageed et al., 2021), trained as a joint 13-label BIO tagger on the 1,125 training rows with the 138 development rows used for epoch selection, widens any overlapping proposal to the encoder boundary, since the overlap half of the score counts characters while an LLM tends to mark only the word it judges wrong. Gemini 3.6 Flash may also widen spans; a disjoint Gemini proposal is admitted only when MARBERTv2 independently predicts an overlapping error. Rows left without a prediction use a fixed fallback chain, and adjacent same-category fragments separated by at most one character are merged. The translation itself is never edited; Appendix A records implementation and prompt details.

4.2 Results and error analysis

The official blind-test ladder records the contribution of each retained stage: raw GPT-5.5 scores 0.4691 Overall, encoder-guided widening reaches 0.4786, extended widening with agreement-gated additions and empty-row rescue reaches 0.4967, and fragment repair produces the final 0.4982 (Table 16). The best fully local alternative combines Gemma-4-31B-IT and MARBERTv2 at character level and reaches 0.4446. In pre-test cross-validation of a direction-routed system, 51.7% of predicted spans match no gold span and 33.7% of gold spans are missed, whereas category accuracy reaches 75.35% once a span is matched (Figure 4). In these predictions, finding and localizing errors is therefore a larger bottleneck than classifying an error after it has been found.

5 Conclusion

We developed first-place systems for all four AlexandriaX-2026 settings, with matched controls identifying where their gains came from. In the constrained track, authentic Libyan and Sudanese supervision outperforms matched duplicate-row controls. Post-evaluation, averaging three full-data models improves spBLEU over every individual seed in all 13 dialects, yielding a stronger single-model recipe without routing. Source-only target-dialect retrieval improves generation without fine-tuning, while error detection improves when semantic proposal and span control are separated.

Across these systems, component value does not simply follow parameter count. Among the submitted candidates, compact E4B is the strongest standalone generator and largest conditional contributor, yet turn-level MBR surpasses every candidate in all 13 dialects. A 162M-parameter encoder likewise improves the boundaries of frontier-model proposals. The authority control separates evidence from generation: reference-only jurors strengthen consensus, but making them selectable forfeits 0.1893 spBLEU. Together, these results support assigning each component its empirically strongest role.

Limitations

The Alexandria dataset study underlying Subtask 1 finds that current models preserve meaning more reliably than they produce authentic local dialect (El Mekki et al., 2026b). Our official results rely on single-reference spBLEU and chrF++, which cannot determine whether a gain reflects greater dialect authenticity or closer agreement with one translator's wording. We did not conduct a separate human evaluation and therefore interpret these scores as benchmark performance rather than direct evidence of naturalness.

Two design choices are specific to the released benchmark. The subdialect mapping depends on patterns in the conversation identifiers, and the constrained fold-in adapter uses the only released LY/SD supervision, leaving no held-out Alexandria split for these dialects. Subtask 2 is also transductive. Because no parallel task data were released, retrieval uses blind-source sentences written in the target dialect to expose vocabulary, morphology, spelling, and style without identifying them as translations. The same pool can nevertheless contain semantically corresponding sentences across dialects. Reciprocal-neighbor filtering removed likely alignments, but a retrospective audit found at least two recoverable counterparts among 3,574 final prompts, with coverage limited to public data. The Subtask 2 result therefore characterizes retrieval-primed inference over this released pool rather than conventional inductive generalization.

For the blind test, unreleased references preclude confidence intervals over official deltas. We therefore report the smallest Subtask 3 increments as observed pipeline stages rather than stable effect estimates. Controls designed after leaderboard feedback are marked post-evaluation and treated as diagnostic; Appendix E.2 reports the Subtask 3 development-to-test analysis.

Finally, the unconstrained Subtask 1 jury and the Subtask 3 proposal pipeline use proprietary API models. Their outputs were cached, but exact regeneration depends on continued access to the reported endpoints.

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A Implementation and Reproducibility Details

Subtask 1 generators. The 12B stage is a QLoRA (Dettmers et al., 2023) SFT of `gemma-4-12b-it` on 97,672 serialized rows (train, AraBench LY/SD with 2,500 and 2,000 rows respectively, public test, and full development): rank 16, alpha 16, dropout 0, learning rate 2×10^{-4} with linear decay, one epoch (3,053 steps) at effective batch 32 (8×4), country-balance temperature $\tau=0.5$, seed 3407, maximum sequence length 1024, and English-only context serialization; CPO follows as in Table 3. The E4B candidate is a bf16 LoRA SFT of `gemma-4-E4B-it` on the roughly 85.4k-row train+AraBench+public-test mixture: rank 32, alpha 64, dropout 0.05, learning rate 2×10^{-4}, two epochs at effective batch 64 (4×16), $\tau=1.0$, seed 42, maximum length 1024, decoded with beam 8 (length penalty 1.0, bf16) over rows with metadata-derived subdialect labels.

The post-evaluation full-data seed and weight-averaging experiment uses a separate recipe from the submitted soup; Appendix C.1 reports its training, merging, and decoding settings alongside all three individual results.

The reference-only Gemma-4-E2B-IT uses the E4B training recipe and data mixture above, with beam-8 decoding. Together with the six API model-prompt variants below, it forms the seven-member jury.

Inference-only components. The six reference-only jurors are `gpt-4.1-mini` (subdialect-named prompt; temperature 0, 300 output tokens), `gpt-5.5` (subdialect-named and full-metadata prompts; reasoning effort none, 800 output tokens), `gemini-3.6-flash` (subdialect-named), and `gemini-3.1-pro-preview` (subdialect-named and full-metadata), the Gemini pair at thinking level low with 512 output tokens and no sampling controls. The Subtask 2 retriever uses `intfloat/multilingual-e5-large` with mean pooling; its translation generator is `unsloth/gemma-4-31b-it`, decoded at temperature 0. The primary Subtask 3 proposer is `gpt-5.5` with 8,000-token completions. `gemini-3.6-flash`, at temperature 0, supplies spans for extended widening, agreement-gated additions, and the empty-row fallback below.

Subtask 3 empty-row fallback. For any row still empty after extended widening and agreement-gated additions, the submitted controller tries, in order: (1) the Gemma-4-31B-IT few-shot extractor combined with MARBERTv2; (2) Gemini 3.6 Flash alone; (3) the supervised-fine-tuned Gemma-4-12B-IT extractor combined with MARBERTv2; and (4) MARBERTv2 alone. Both local hybrids use span union with widest-span merging of overlapping predictions. The controller takes the first nonempty span set, then merges adjacent same-category fragments.

A.1 Prompt contracts

The following schemas record the prompt choices tied to measured system behavior. Bracketed text denotes inserted data; provider-specific request parameters are reported above.

Subtask 1 reference-only jurors. Four model-prompt variants, `gpt-4.1-mini`, `gpt-5.5`, `gemini-3.6-flash`, and `gemini-3.1-pro-preview`, use the same subdialect-named textual contract through provider-specific message wrappers. Its system instruction is:

```
You are a professional, precise, and
native-level translator. Your task is
to translate the provided text from the
source language into the specified
target language. Output only the best
possible translation, with no
explanations, alternatives, or
additional text. Preserve formatting,
punctuation, and any untranslatable
elements (such as names or technical
terms) as appropriate.
```

The user message is:

```
Translate the following text into
[derived subdialect], only print out
the translation, not add any
explanations: [English turn]
```

The `gpt-5.5` and `gemini-3.1-pro-preview` full-metadata variants additionally expose the same released fields, with minor serializer-specific wording and formatting. The GPT-5.5 full-metadata system instruction is:

```
You are a native-speaker translator who
produces authentic spoken dialectal
Arabic (NOT Modern Standard Arabic).
Match the exact local dialect
requested. Preserve speaker/addressee
gender agreement. Keep conventional
code-switched loanwords, names,
numbers, URLs, and emojis as-is. Output
ONLY the translation, no quotes, no
explanations, no transliteration.
```

System	Data	Adaptation or retrieval	Inference and role
ST1 unconstrained, 12B CPO	Alexandria train, development, and public test; AraBench (2,500 LY + 2,000 SD examples) (Sajjad et al., 2020)	Gemma-4-12B-IT; 13,049 CPO pairs; $\beta = 0.1$; SFT anchor 1.0; lr 10^{-5}; one epoch; effective batch 16; seed 3407	selectable local candidate; metadata conditioning; terminal-punctuation normalization
ST1 unconstrained, E4B	Alexandria train and public test plus AraBench LY/SD; no development exposure	Gemma-4-E4B-IT LoRA SFT	beam-8; selectable local candidate
ST1 constrained	train+development soup; separate train+public-test adapter routed only for LY/SD	one NileChat-3B-Base backbone; three rank-32, alpha-64 soup deltas plus one rank-32, alpha-64 fold-in delta	country-selected LoRA route; derived subdialect label; beam-8; self-context; MR/MA k=3 demonstrations
ST2	final retrieval uses released blind sources; ArBanking77 used only for exploratory retriever selection; no parallel task training data	multilingual E5-Large; reciprocal 32-neighbor filtering; margin ranking; $k = 32$	Gemma-4-31B-IT; greedy decoding; 32k context; guided generation; exact-copy regeneration
ST3	1,125 training rows; 138 development rows for epoch and system selection	`UBC-NLP/MARBERTv2` joint 13-label BIO; direction as first segment; lr 2×10^{-5}; batch 16; max length 256; seed 7; up to 25 epochs	GPT-5.5 proposer; MARBERTv2 boundary control; agreement-gated Gemini 3.6 Flash additions; empty-row rescue; gap-1 same-category merge

Table 3: Final-system implementation summary. Selectable ST1 candidates are summarized in Table 2; reference-only jurors and their prompts are described in Appendix A.

The Gemini full-metadata variant uses this shorter instruction, concatenated before the user text rather than sent as a separate system message:

```
You are a native-speaker translator who
produces authentic spoken dialectal
Arabic (NOT Modern Standard Arabic).
Match the exact local dialect
requested. Preserve speaker/addressee
gender agreement. Keep code-switched
loanwords, names, numbers as-is. Output
ONLY the translation.
```

The full-metadata user-message schema is:

```
Target dialect: [derived subdialect]
([country]).
Domain: [domain].
Speaker: [speaker] -> Addressee:
[addressee].
Gender direction (speaker ->
addressee): [direction].
Prior dialogue turns: [preceding
English turns].
Now translate ONLY this turn into
authentic [derived subdialect]:
[English turn]
```

No reference Arabic history is supplied. The Gemini metadata serializer retains an empty country-tagged Arabic line after each preceding English turn; GPT-5.5 includes only the English history lines. Conversation identifiers are not inserted directly; they are used only to derive the subdialect label.

Subtask 1 constrained completion. Training and inference use the same completion frame:

```
You are an expert translator. Translate
the English sentence into [derived
subdialect].
### Metadata: [country, domain,
participants, speaker, direction]
### Conversation History: [preceding
English turns and Arabic translations]
### Sentence to Translate: [English
turn]
### Translation:
```

Each training example is one English turn in this frame, with the preceding turns of its conversation serialized as history; during inference, the Arabic history contains the model's own preceding outputs. For MR and MA, three complete training instances serialized in this frame precede the query, preserving the format used during fine-tuning.

Subtask 2 retrieval-guided generation. The prompt names the source and target dialects, inserts the retrieved sentences, and explicitly defines their role:

```
The following are authentic [target]
banking usage examples retrieved only
from the supplied source sentences.
They are not translations of the input.
Use only their [target] vocabulary,
morphology, spelling, and style. Do not
copy their meaning or transaction
details.
```

The retained generation instruction then asks the model to consider internally (1) a conservative meaning-preserving rendering, (2) a natural target-dialect rendering, and (3) a rendering aligned with the retrieved usage. It returns only one synthesized sentence, without alternatives, analysis, or scores.

Subtask 3 tagged span annotation. Four training examples precede the query, ranked first by matching translation direction and then by Arabic token overlap; at least one example covers a rare LQM category. The proposer then receives the direction, English source, and Arabic MT, followed by `Tagged:`. Its output contract is to copy the Arabic MT verbatim and wrap each minimal error span as `<category>span</category>`, where `category` is one of the six LQM labels. The instruction resolves the dominant ambiguity by assigning a meaning-preserving variety or register mismatch to sociolinguistics, and a meaning change to semantics. Only the tagged Arabic copy is returned. We extract the tagged substrings and locate them in the original MT, so offsets are reconstructed rather than generated numerically.

B Official Score Details

Table 1 reports the aggregate blind-test results. For Subtask 3, Overall is the mean of overlap-span F1 (0.5426) and pooled error-class F1 (0.4538); exact-span F1 is 0.2319 and does not enter Overall. For reproducibility, we follow the deployed scorer, whose class component pools overlap-matched spans before computing F1. Our development implementation reproduces the deployed score.

Score differences are computed before rounding whenever full-precision scorer records are available; the Subtask 3 comparison record uses the preserved four-decimal scores.

Table 4 reports the two officially submitted Subtask 1 systems by country. The unconstrained system scores higher in all 13 countries and on both metrics. Both systems score highest on Syrian Arabic and lowest on Mauritanian Arabic.

	Constrained*		Unconstrained†	
Country	spBLEU ↑	chrF++ ↑	spBLEU ↑	chrF++ ↑
EG	33.07	47.36	37.54	51.37
JO	36.17	50.01	38.80	52.79
LB	32.44	46.97	35.75	49.56
LY	26.72	43.43	29.45	46.17
MA	24.14	40.57	27.57	42.90
MR	17.67	34.83	20.18	36.58
OM	35.51	49.41	38.10	51.57
PS	33.22	47.64	36.04	50.15
SA	32.86	47.47	36.95	51.06
SD	28.17	43.10	30.32	45.25
SY	39.75	53.77	42.92	56.52
TN	29.41	43.55	33.16	47.16
YE	26.37	42.42	28.57	44.64
Macro	30.4233	45.4250	33.4891	48.1320

Table 4: Per-country blind-test scores for our two first-place official Subtask 1 submissions. *Country-routed NileChat (30.4233 macro spBLEU): LY and SD use the train+public-test adapter; the other 11 countries use the train+development soup. †Candidate-constrained MBR (33.4891), with the four candidates in Table 2. Neither column includes post-evaluation changes. Codes: EG Egypt, JO Jordan, LB Lebanon, LY Libya, MA Morocco, MR Mauritania, OM Oman, PS Palestine, SA Saudi Arabia, SD Sudan, SY Syria, TN Tunisia, YE Yemen. Per-country values are rounded to two decimals; macro scores are reported by the scorer.

C Subtask 1: Extended Experiments and Analysis

C.1 Constrained-track ablations

Authentic supervision versus additional updates. Table 5 compares five seed-42 `NileChat-3B-Base` adapters trained with the same LoRA recipe and decoded identically. At two added-data budgets, duplicating existing rows controls for both the number of examples and optimizer updates. Authentic public-test supervision outperforms its matched duplicate-row control by 1.0208 macro spBLEU when only LY/SD rows are added and by 1.6250 when the complete public-test split is added. The gain therefore follows new labeled content rather than a longer optimization schedule.

Seed variation and weight averaging. We repeat the full-data recipe with seeds 42, 123, and 7 (Table 6). Each `NileChat-3B-Base` run uses all 93,172 training, development, and public-test examples: LoRA rank 32, alpha 64, dropout 0.05, attention and MLP projections, AdamW at 2×10^{-4}, bf16, two epochs, and effective batch 16 (11,648 optimizer steps). Only the random and data-order seed changes. Training uses gold dialogue history.

Training mixture	Macro spBLEU	Δ	LY	SD	Mean, other 11
Training + development	29.2323	baseline	20.9637	21.4406	30.6923
Added-data budget matched to the public LY/SD subset					
Duplicated training rows	29.0971	-0.1352	20.8941	21.7975	30.5064
Authentic public LY/SD rows	30.1179	+0.8856	26.7867	**27.8097**	30.6305
Added-data budget matched to the complete public-test split					
Duplicated training rows	28.8483	-0.3840	21.7299	21.2975	30.1819
Authentic complete public test	**30.4733**	**+1.2410**	**27.2953**	27.5768	**31.0256**

Table 5: Unofficial post-evaluation supervision controls produced by the official scorer on the blind test. All rows use the same seed-42 adapter recipe and inference configuration. Δ is relative to training on the training and development splits. The duplicate rows contain no new translations. Bold marks the highest value in each numeric column.

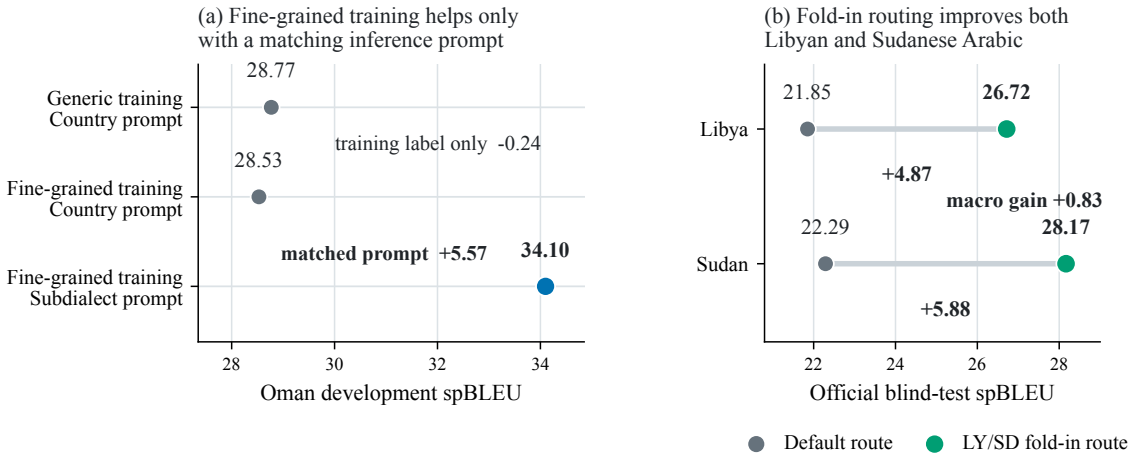


Figure 2: Subtask 1 conditioning and constrained routing controls. Panel (a) compares two Omani-only `Gemma-4-12B-IT` LoRA adapters trained on 6,280 rows and evaluated on 1,109 held-out Omani development turns. The adapters share the base model, recipe, and seed; only the country-level versus subdialect-level training label changes. The subdialect-trained adapter is evaluated under both country-level and matched subdialect prompts. Panel (b) shows the official blind-test changes from routing LY/SD to the fold-in adapter.

At inference, all arms use beam-8, derived subdialect labels, and generated Arabic self-context, without a special LY/SD hint.

Following weight-space averaging (Wortsman et al., 2022), we merge each LoRA adapter into the same backbone, average the three complete models uniformly in fp32, and save one bf16 model. We do not average the LoRA factors directly or combine predictions at inference. This requires three training runs but retains one 3.086B-parameter model for decoding. The two additional seeds and equal-weight soup were fixed before their post-evaluation scores were obtained.

The individual seeds have mean macro spBLEU 30.4537 and sample standard deviation 0.0171. Their soup reaches 30.9726, improving on the mean by 0.5189 and on the best seed by 0.4993. It exceeds the strongest seed within each of the 13 dialects, with margins from 0.0131 in YE to 0.6433 in MR. Macro chrF++ also improves, by 0.3884 over the strongest individual model on that metric. These results support weight averaging for this full-data recipe; the three-seed dispersion is descriptive, not a confidence interval or significance test. Appendix C.4 reports a separate post-evaluation experiment replacing the submitted unconstrained system's NileChat candidate with this soup.

Decoding. For the full-data seed-42 adapter, greedy decoding scores 26.6855, beam-4 scores 30.3822, and beam-8 scores 30.4733. Beam search supplies nearly all of the 3.7878-point gain over greedy decoding, while increasing the width from four to eight adds 0.0912. On the 11-country development set, beam-8 exceeds beam-5 by 0.121 macro spBLEU, with a 95% conversation-bootstrap interval of $[+0.025, +0.222]$.

Fine-grained labels and generated dialogue context. Table 7 isolates two inference choices in one fixed system: the seed-42 `NileChat-3B-Base`

Full-data model	Macro spBLEU	Macro chrF++
Seed 42	30.4733	45.4750
Seed 123	30.4426	45.4991
Seed 7	30.4451	45.4739
Equal-weight soup	**30.9726**	**45.8875**

Table 6: Post-evaluation full-data seed comparison, using the official scorer on the blind test. The three seeds share one training recipe; all four arms use beam search with width 8 and sampling disabled. The soup averages the three merged models with equal weights and is decoded independently. It is not the train+development soup used in the submitted system. Bold identifies the soup.

Prompt label	English history only	+ Generated Arabic history
Country-level	28.6890	28.8358
Derived subdialect	30.2787	**30.4733**

Table 7: Unofficial post-evaluation factorial for the full-data seed-42 NileChat-3B-Base adapter, scored on the official blind test. All arms retain English source history; Arabic history contains prior model outputs only. Bold marks the best configuration.

LoRA adapter trained on all released training, development, and public-test data and decoded in bf16 with beam width 8. First, the prompt names either the released country or the subdialect derived from the conversation-prefix mapping in Section 2.1. Second, target-side dialogue history is either omitted or generated by the model itself. Every arm retains the preceding English turns. With generated self-context, the conversation is translated in order and the model's own preceding Arabic translations, never reference translations, are supplied when translating the next turn. Averaged over the other factor, the derived label contributes 1.6136 macro spBLEU, generated self-context contributes 0.1707, and their interaction, the difference between the two simple effects, is 0.0478. Their effects are close to additive in this fixed system.

Why label granularity matters. Figure 2(a) isolates the interaction between training-label and test-prompt granularity. We fine-tune two Omani-only `Gemma-4-12B-IT` LoRA adapters on the same 6,280 training rows and evaluate them on the same 1,109 held-out Omani development turns. The base model, LoRA recipe, and seed are identical; only the training prompt differs. One adapter is trained with the generic label *Omani Arabic*, while the other receives the annotated Al-Wafi, Rustaqi, or Suri label for each example. At inference, the generic adapter with a country prompt scores 28.77. The subdialect-trained adapter scores 28.53 with that same coarse prompt, but 34.10 when given the matching subdialect, a gain of 5.57. Fine-grained supervision therefore helps only when its label granularity is preserved at inference.

The controlled result explains the broader near-matched official comparison in Section 2.1. The five countries containing multiple annotated subdialects contribute +1.14 to the 13-country macro average (OM +7.47, SY +2.73, YE +2.13, PS +1.43, and SA +1.01); the other eight contribute only +0.04. Lexical analysis suggests a plausible mechanism: Omani subdialects, for example, use different forms for common functions such as *want*. Across the five multi-subdialect countries, lexical separability and the official gain have a positive exploratory correlation ($r = 0.75$); the sample is too small for a general claim.

Figure 2(b) shows the complementary data intervention. Routing LY and SD to the fold-in adapter raises their official spBLEU by 4.87 and 5.88, respectively, while the other 11 country outputs remain byte-identical.

C.2 Model landscape and translation behavior

Model landscape. Table 8 summarizes our broader model study. The first two blocks share one common full-development evaluation; the remaining blocks record exploratory screens or dialect-specific slices. On the common evaluation, Gemma-4-E4B-IT rises from 23.49 zero-shot to 33.58 after fold-in SFT and beam search. The corresponding official-test system scores 32.2336, making the compact E4B our strongest standalone submitted candidate. Excluding the daggered training-overlap diagnostic, it also exceeds every tested 12B and 31B configuration in the common evaluation, showing that raw parameter count alone does not determine performance here.

The k=32 Gemma-4-31B-IT row first restricts demonstrations to the query's released conversation-identifier group and then uses BM25 to fill any remaining slots. It is included to document the tested retrieval strategy, not as an independent training recipe. The daggered CPO row is likewise a diagnostic because its evaluation data were included in its training mixture. The NLLB and Seed-X-PPO model families are described by NLLB Team et al. (2022) and Cheng et al. (2025),

Model	Adaptation and decoding	Evaluation set	spBLEU
Common full-development evaluation: prompted systems, 12,250 turns, 11 countries			
Gemma-3-1B-IT	zero-shot, greedy	full development	3.77
Gemma-3-4B-IT	zero-shot, greedy	full development	10.50
NileChat-3B (untuned)	zero-shot, greedy	full development	13.11
Gemma-4-E2B-IT	zero-shot, greedy	full development	21.10
Gemma-4-E4B-IT	zero-shot, greedy	full development	23.49
Gemini 2.5 Pro	zero-shot, prompt-matched	full development	23.93
Gemini 3.6 Flash	zero-shot, prompt-matched	full development	24.04
Gemma-4-31B-IT	zero-shot, greedy	full development	24.93
Gemma-4-31B-IT	BM25 few-shot, k=16	full development	27.93
Gemma-4-31B-IT	identifier-group BM25 few-shot, k=32	full development	30.07
Common full-development evaluation (continued): fine-tuned systems			
Gemma-3-4B (base)	LoRA SFT, beam-8	full development	29.60
Gemma-3-4B-IT	LoRA SFT, beam-8	full development	29.82
NileChat-3B-Base	constrained LoRA SFT, beam-8	full development	29.86
Qwen3.5-9B	fold-in LoRA SFT, beam-5	full development	30.72
Gemma-4-E2B-IT	LoRA SFT, beam-8	full development	31.71
Gemma-4-12B-IT	LoRA SFT, train only, greedy	full development	31.95
Gemma-4-12B-IT	+ LY/SD fold-in and DPO, greedy	full development	32.06
Gemma-4-E2B-IT	fold-in SFT, beam-8	full development	32.20
Gemma-4-12B-IT	+ CPO, MR/TN routing, greedy	full development	32.63
Gemma-4-E4B-IT	LoRA SFT, beam-8	full development	32.99
Gemma-4-E4B-IT	fold-in SFT, beam-8	full development	**33.58**
Gemma-4-12B-IT	fold-in SFT + CPO	development used in training	34.13†
Exploratory samples, screens, and holdout folds			
Hy-MT2-1.8B	zero-shot, greedy	110-turn sample	9.62
Granite-4.0-H-3B-AR	zero-shot, greedy	55-turn sample	12.68
Qwen3.5-9B	zero-shot, greedy	440-turn sample	13.25
Command-R7B-Arabic	zero-shot, greedy	440-turn sample	16.91
GPT-5	zero-shot	440-turn sample	19.85
Seed-X-PPO-7B	LoRA SFT, 0.36 epoch, beam-3	440-turn sample	21.40
Gemma-4-12B-IT	zero-shot, greedy	1,100-turn screen	24.04
Qwen3.5-9B	LoRA SFT, beam-5	2,442-turn holdout fold	31.18
Dialect-specific slices of the development set			
DarijaTranslation-V1	off the shelf	MA column	5.63
Masrawy v2 (oddadmix)	off the shelf	EG column	13.77
NLLB-200-1.3B	MA-only SFT	MA column	14.79
Atlas-Chat-2B	MA-only LoRA SFT	MA column	14.98
Masrawy v2.9 (NAMAA)	off the shelf	EG column	15.14
NLLB + AraDiCE	two-stage EN to MSA to dialect	5 columns, 200 turns each	16.23
50M-Egyptian-v1	off the shelf, own template	EG column	18.28
English2Dialect	off the shelf	5-column mean	20.03
Levanti	off the shelf	4-column mean	23.86

Table 8: Subtask 1 model landscape, separated by evaluation protocol. Rows in the first two common full-development blocks are directly comparable except for the daggered diagnostic, which includes the development set in training. Other blocks report only the stated sample, fold, or dialect slice. Here, fold-in denotes an expanded mixture that adds released public-test supervision and, where used by the experiment, AraBench LY/SD rows. Bold marks the strongest held-out full-development result. DPO: direct preference optimization.

respectively.

What remains beyond mechanical normalization. One simple surface intervention has a measurable but limited effect: on the 12B CPO candidate, deterministic terminal-punctuation normalization raises official blind-test macro spBLEU from 31.7873 to 32.0318, a gain of 0.2446. Separately, Table 9 analyzes a 12B CPO checkpoint preceding the submitted candidate on the 12,250-turn development set covering 11 countries, using a layered word-level taxonomy. Exact matches account for 50.0% of reference words, while spelling and clitic-level morphology together account for under 10%. The largest non-exact-match category, at 39.2%, contains different surface forms after normalization that string matching alone cannot separate into valid paraphrases and mistranslations.

Additional probes find little recoverable mass in simple transformations. Dialect-cluster leakage accounts for 0.2%; the prediction-to-reference length ratio is 1.06, giving no evidence of systematic shortening; and all numeric tokens form only 0.17% of reference words, with an oracle ceiling of +0.03 macro spBLEU. Normalizing both predictions and references for spelling and morphology changes macro spBLEU by +1.59, with gains in MR (+12.6) and YE (+11.3) offset by losses including EG (-7.2) and MA (-6.1). This diagnostic changes the scoring inputs on both sides; it is not an achievable post-processing gain. Manual inspection of the MR cases reveals many synonym pairs, such as *lāzim* and *ḍarūrī* for "must" or "necessary." For MR, the roughly 30% exact-match and 60% different-word profile is consistent with substantial single-reference variation, but source-grounded native-speaker annotation would be required to distinguish that variation from genuine translation errors.

Alignment outcome	Reference words (%)
Exact match	50.0
Spelling variant	1.8
Clitic-level morphology	7.9
Different surface form after normalization	39.2
Code-switch	∼1.0
Unmatched numeral	∼0.1

Table 9: Word-level alignment outcomes for the earlier 12B CPO development checkpoint after layered normalization. Percentages use all reference words as the denominator; code-switch and numeral rows count only words still unmatched. The morphology class contains a matching stem with a different clitic.

C.3 Selection granularity and utility

Selection granularity. The submitted selector changes systems on 71.4% of consecutive turns. A switching penalty of $\lambda = 2$ lowers the rate to 54.1% but costs 0.0850 macro spBLEU. Selecting one system for an entire conversation costs 0.7391, although it remains 0.5163 above the best standalone candidate. Under single-reference spBLEU, turn-level complementarity is therefore more valuable than enforcing system consistency across a conversation.

Utility alignment. Table 10 freezes the pool, roles, deduplication, and restricted argmax while changing only the pairwise utility. Sentence-chrF++ improves official chrF++ by 0.1686 but gives up 0.4501 spBLEU. COMET-22 (Rei et al., 2022) changes 60.2% of selections, largely toward the NileChat soup, and falls below the strongest standalone candidate on both official metrics. Under this fixed pool, each string utility favors its corresponding metric, while the COMET-22 utility improves neither official string metric. This experiment complements semantic, efficient, and source-based MBR variants (Zhang et al., 2022; Trabelsi et al., 2024; Lyu et al., 2025).

C.4 Candidate, jury, and eligibility contributions

Jury-family interaction. The two official submissions and two post-evaluation controls form a complete 2×2 factorial over the open E2B juror and the six API jurors (Table 11). E2B contributes 0.1025 without the API jurors and 0.1703 with them; the API family contributes 0.2263 without E2B and 0.2940 with it. Their positive interaction, computed with the same difference-of-differences

Pairwise utility	Changed (%)	spBLEU	chrF++
Sentence-spBLEU	baseline	**33.4891**	48.1320
Sentence-chrF++	23.4	33.0389	**48.3006**
COMET-22	60.2	32.0266	47.0838

Table 10: Utility comparison under a fixed candidate pool and output-eligibility rule. Changed selections are measured against the submitted sentence-spBLEU utility. The alternatives are unofficial post-evaluation controls. Bold marks the best score for each metric.

Reference-only jury	No API jurors	Six API jurors
No E2B	33.0925 (post)	33.3188 (post)
E2B	33.1950	**33.4891**

Table 11: Jury-family factorial with the four selectable candidates and fixed sentence-spBLEU utility. Parentheses mark post-evaluation controls; the other two cells are official submissions. Bold marks the highest score.

convention, is 0.0677, and the API family improves 10 of 13 dialects in either E2B condition.

Juror stability. Table 12 compares all eight tested reference-only jurors on two tuning splits. Among the seven retained jurors, only the E2B system, the GPT-5.5 target-dialect prompt, and the Gemini 3.6 target-dialect prompt have positive leave-one-out effects on both splits. Adding the excluded GPT-5.4 mini back to the final pool lowers scores on both splits. Exact development Shapley values over all 256 coalitions of the eight tested jurors sum to that diagnostic pool's +0.4329 improvement over four-candidate MBR. The signs differ across splits and attribution methods, so these analyses diagnose juror behavior rather than define a tuned subset.

Candidate contributions. In post-evaluation controls, demoting one local candidate to reference-only while retaining its reference hypotheses lowers macro spBLEU by 0.1997 for 12B CPO, 0.1332 for 12B CPO + ICL, 0.4175 for E4B, and 0.1366 for the NileChat soup. Each candidate therefore makes a measurable conditional contribution under the fixed selector, with E4B contributing most strongly and helping all 13 countries.

Table 13 shows a second form of complementarity. The best complete candidate varies by dialect: ICL leads in seven, E4B in four, and 12B CPO in two. Although the NileChat soup leads none, it supplies 12.6% of selected turns. MBR beats every dialect's strongest complete candidate, from +0.09 in LY to +1.51 in MR, and exceeds a retrospective dialect-level system oracle, which selects one complete candidate per dialect using blind-test scores, by 0.8145. Its gains therefore arise within dialects rather than only from coarse routing.

Reference evidence versus output eligibility. Figure 3 distinguishes candidate combination, added jury evidence, and output eligibility. Combining the four candidates raises macro spBLEU from the strongest standalone score of 32.2336 to 33.0925. Adding reference-only jurors then raises it to 33.1950 with E2B and 33.4891 with the six API jurors. A matched post-evaluation control keeps the full 11-system pool, utility, and deduplication fixed but allows every system to supply the output; it scores 33.2998, forfeiting 0.1893 of the restricted selector's gain. Juror evidence helps under both eligibility regimes, but restricting outputs to the four validated local candidates preserves more of that benefit.

When all 11 systems are eligible, jurors provide 4,893 of 14,459 outputs. Restriction improves 8 of 13 countries, led by OM +1.59, YE +0.72, and LB +0.50; SD is the largest exception, with unrestricted selection ahead by 0.81. The unrestricted control still exceeds both four-candidate MBR and the E2B-only jury, confirming that the result concerns the degree of benefit rather than whether juror evidence helps at all.

Post-evaluation candidate replacement. We replace the original NileChat candidate with the full-data soup from Appendix C.1, keeping the other three candidates, seven reference-only jurors, and selection procedure unchanged. Macro spBLEU rises from 33.4891 to 33.5246 and chrF++ from 48.1320 to 48.1811. The replacement is selected on 14.8% of turns, compared with 12.6% for the original candidate, and spBLEU improves in eight of thirteen dialects. Because selectable candidates also supply reference hypotheses, this comparison measures the combined effect of replacing an output option and its reference evidence.

D Subtask 2: Retrieval Details and Integrity

D.1 Retrieval configuration

For a query q from dialect A to dialect B, the candidate bank contains the unique released blind-test sources written in B. We encode the bank with multilingual E5-Large using mean pooling and rank

Reference-only juror	Development effect	Public-test effect	Development Shapley
Gemma E2B, fine-tuned	+0.020	+0.771	-0.018
GPT-4.1 mini, dialect prompt	+0.161	-0.025	+0.086
GPT-5.4 mini, dialect prompt	-0.068	-0.086	-0.062
GPT-5.5, dialect prompt	+0.144	+0.018	+0.086
GPT-5.5, metadata prompt	+0.063	-0.042	+0.093
Gemini 3.6 Flash, dialect prompt	+0.023	+0.012	+0.062
Gemini 3.1 Pro, dialect prompt	+0.200	-0.072	+0.085
Gemini 3.1 Pro, metadata prompt	+0.156	-0.031	+0.101

Table 12: Juror effects across splits and attribution methods. For the seven retained jurors, the first two numeric columns report the final-pool score minus the leave-one-out score. The excluded GPT-5.4 mini row instead reports its add-back score minus the final-pool score. Seven of eight development effects are positive, compared with three on public test. The third column gives exact development Shapley values over all 256 coalitions of the eight tested jurors, including GPT-5.4 mini; these sum to the eight-juror diagnostic pool's +0.4329 gain. Positive values indicate a beneficial contribution.

	selectable candidates (standalone)				selector	
Dialect	12B CPO	12B ICL	E4B	NileChat	MBR	Δ
EG	35.69	**36.91**	35.62	33.07	37.54	+0.63
JO	37.32	**38.39**	37.67	36.17	38.80	+0.41
LB	33.48	**34.83**	33.80	32.44	35.75	+0.93
LY	29.05	27.90	**29.37**	21.85	29.45	+0.09
MA	**26.47**	26.38	25.69	24.14	27.57	+1.10
MR	18.06	**18.67**	18.49	17.67	20.18	+1.51
OM	36.00	35.98	**37.92**	35.51	38.10	+0.18
PS	34.86	33.83	**35.32**	33.22	36.04	+0.73
SA	35.34	**35.86**	35.54	32.86	36.95	+1.09
SD	29.61	29.11	**29.61**	22.29	30.32	+0.71
SY	**41.78**	41.41	41.66	39.75	42.92	+1.14
TN	31.31	**32.12**	30.86	29.41	33.16	+1.04
YE	27.44	**27.53**	27.49	26.37	28.57	+1.04
Macro	32.0318	32.2235	**32.2336**	29.5969	33.4891	+1.2554
Retrospective dialect-level system oracle		32.6746			33.4891	+0.8145

Table 13: Official blind-test spBLEU for the four selectable candidates and the submitted MBR selector. Bold marks the strongest complete candidate in each dialect; Δ is MBR minus that candidate before rounding. The final row retrospectively chooses one complete candidate per dialect using blind-test scores and was not used in system development.

each candidate c by the relative margin

$$m(q, c) = \frac{2\cos(q, c)}{\bar{s}_4(q, B) + \bar{s}_4(c, A)}, \qquad (2)$$

where $\bar{s}_4(q, B)$ is the mean similarity between q and its four nearest sources in B, and $\bar{s}_4(c, A)$ is defined symmetrically over sources in A. Before ranking, we discard c if q occurs among its 32 nearest dialect-A sources. This reciprocal filter reduces the chance of retrieving a cross-dialect counterpart using only source-side similarities. The top k survivors are supplied as unpaired examples of target-dialect usage, without cross-dialect alignment. This resembles monolingual target-side retrieval for MT (Bouthors et al., 2025), but the bank here consists only of released source messages.

Generation uses greedy decoding from Gemma-4-31B-IT. The submitted prompt asks the model to weigh three renderings internally: one preserving the source closely, one favoring natural target-dialect expression, and one aligned with the retrieved usage examples. This is a single-call latent variant inspired by reasoning prompts for MT, not explicit tree search (Nguyen and Xu, 2025). It returns only one synthesized translation. An exact-copy guard regenerates an output when its normalized text copies an active example; persistent copies use the Gemma-4-31B-IT zero-shot output produced without retrieval.

D.2 Official system ladder

Table 14 reports the complete official blind-test ladder. Scores increase at every submitted retrieval depth. The k=8 system sends copied outputs

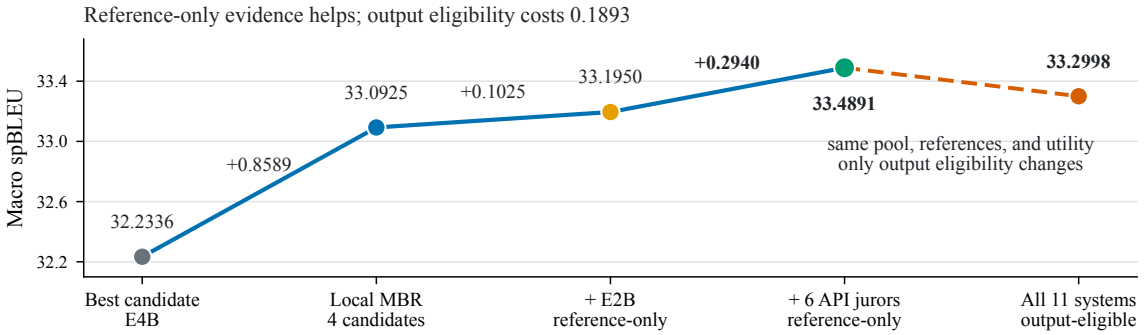


Figure 3: Candidate combination, reference evidence, and output eligibility have different effects. The horizontal path first combines the four candidates and then adds reference-only evidence. The final branch keeps the complete pool and scoring rule fixed but grants every juror output eligibility; this post-evaluation control scores 0.1893 below the submitted restricted selector.

directly to the no-retrieval fallback, whereas the k=16 and k=32 systems first remove the copied example, promote the next-ranked candidate, and regenerate. Consequently, k=16 to k=32 is the controlled depth comparison. With k=32 retrieval and repair fixed, guided generation (latent deliberation) adds 0.45 spBLEU (27.9496 to 28.4011).

System	spBLEU	chrF++
Copy source unchanged	12.81	27.31
Gemma-4-12B-IT zero-shot	23.7949	37.8899
Gemma-4-31B-IT zero-shot	24.6898	38.8862
Retrieval, k=8, direct fallback	26.8595	40.7924
Retrieval, k=16, regenerate	27.4687	41.2666
Retrieval, k=32, regenerate	27.9496	41.7926
k=32 **+ guided generation**	**28.4011**	**42.1509**

Table 14: Official Subtask 2 blind-test ladder. Scores use FLORES-200-tokenized spBLEU and chrF++; the submitted system is bold.

D.3 Development provenance and retrospective audit

Subtask 2 provided no dedicated training or development split. We used public ArBanking77 data (Jarrar et al., 2023) only for exploratory retriever selection. After finding that some rows align with the benchmark, we excluded measurements on those rows from independent performance claims. The final retrieval bank and every headline comparison instead use the released blind-test sources and official blind-test scores, respectively; generation does not query ArBanking77, hidden references, intent labels, or identifiers linking corresponding rows.

Table 15 separates the exact-copy handling performed during generation from the later counterpart audit. In the submitted run, regeneration removed the copied example and promoted the next-ranked candidate; persistent copies used the no-retrieval Gemma-4-31B-IT output.

Event	Incidence
Outputs initially copying an active example	30/3,574 (0.84%)
Non-copy after regeneration	26/30 flagged (86.7%)
No-retrieval fallback	4/3,574 (0.11%)
Known counterpart collisions, k=8 prompts	0/3,574
Known counterpart collisions, final k=32 prompts	2/3,574 (0.06%)
Outputs reusing counterpart wording	1/3,574 (0.03%)

Table 15: Copy handling and retrospective counterpart audit. Rates use all 3,574 outputs or prompts unless marked as conditional on a flagged output. Counterpart counts include only exact matches recoverable from public data.

Only prompts with recoverable public counterparts could be audited, so two is a lower bound on exact counterpart exposure and the audit cannot establish semantic independence. The reciprocal filter and output guard reduce counterpart exposure and verbatim copying, but they do not detect all semantic overlap.

E Subtask 3: Controller Analysis and Error Diagnosis

E.1 Task calibration and scored systems

The released labels are strongly imbalanced: sociolinguistic errors account for 57.9% of the 2,240 training and development spans, followed by semantic errors at 24.4%. On the 138-row development set, a trivial no-model baseline that marks the entire Arabic translation as a sociolinguistic error scores 0.3687 Overall, above the organizers' NileChat-3B baseline at 0.2612. Its exact-span F1 is only 0.0474, however. The result exposes an

important property of the task: broad character coverage and the majority category can produce a deceptively strong score, while useful systems must also localize errors precisely.

Table 16 lists our officially scored blind-test systems. The submitted controller starts from GPT-5.5 semantic proposals, applies MARBERTv2 boundary control, admits disjoint Gemini proposals only with independent encoder support, rescues empty rows through a fixed fallback chain, and merges adjacent same-category fragments. Figure 4 summarizes the resulting score progression and the pre-test error diagnosis.

The four-system character-probability combination instead pools GPT-5.5, MARBERTv2, Gemini 3.6 Flash, and the Gemma-4-12B-IT SFT extractor. It retains characters covered by at least two systems, joins contiguous retained characters into spans, assigns each span a category by voting among overlapping predictions, and merges adjacent same-category fragments. This alternative scores 0.4825, below the submitted controller's 0.4982.

E.2 Validation transfer

Pre-test decisions used stratified five-fold cross-validation over all 1,263 released training and development rows, with tagger retraining and fold-aware demonstration selection in every fold. The fully local Gemma-4-31B-IT/MARBERTv2 span-union hybrid averaged 0.4549 Overall across folds (standard deviation 0.0163); pooled out-of-fold scoring gave 0.4573.

Transfer depended strongly on how a change was validated. A character-level union measured against the exact span-union baseline it replaced improved cross-validation by 0.0084 (95% CI [+0.0049, +0.0123]) and retained a +0.0037 blind-test gain. On the development configuration used to select the submitted extension (cached GPT-5.5 proposals and selected MARBERTv2 tagger), extended widening and agreement-gated additions raised Overall from 0.4374 to 0.4567 (+0.0193). The configuration differs from Table 17's metric-comparison diagnostic. The submitted extension, which also included empty-row rescue, improved blind-test Overall by 0.0181. Fragment merging shrank from +0.0069 in cross-validation to +0.0015 on the blind test. By contrast, changes selected on the 138-row development set or direction-level slices of roughly 27 rows often reversed. These results favor matched comparisons on the largest available labeled split over selecting small global gains from a single development partition.

E.3 Why span discovery remains the bottleneck

Cross-model agreement separates gold-matched from spurious proposed spans at only 0.70 to 0.74 area under the ROC curve (AUC) on the offline splits, which is too weak for standalone filtering. We therefore use agreement only to gate disjoint additions. Two post-evaluation controls confirm that broader consensus does not recover the missing evidence: extending submitted spans to character regions supported by at least two other systems scores 0.4951, and span-set MBR over all eleven scored systems reaches 0.4897, both below the submitted 0.4982. Consensus can adjudicate among proposed spans, but it cannot recover an error region that no component proposed.

Metric sensitivity. The deployed scorer averages character-overlap F1 and pooled class F1. We also re-score the controller stages with partial-overlap, partial-credit matching and micro-averaging (MPP-micro), a recommended robust span-evaluation regime (Perrella et al., 2026). Table 17 shows that agreement-gated additions and fragment merging retain positive gains under MPP-micro. Boundary widening is more metric-dependent: its cross-validation gain falls from +0.0090 under the deployed scorer to +0.0019 under MPP-micro because added characters improve mask coverage but reduce pair precision. The diagnosis itself is stable: unmatched and missed spans remain substantially more common than wrong categories among matched spans.

E.4 Error analysis

Table 18 analyzes GPT-5.5 proposals with MARBERTv2 boundary widening on the 1,263 cross-validation rows, rather than the direction-routed system in Figure 4(b). Of 2,240 gold spans, 772 receive no overlapping prediction; 1,773 of 3,241 predictions match no gold span. Once a span is matched, its category is correct in 1,100 of 1,468 cases (74.9%). The remaining confusions concentrate in the two head categories: 247 of 368 matched errors are assigned semantics or sociolinguistics. This again identifies span discovery as the larger bottleneck.

System	API models	Overall	Overlap F1	Class F1
Submitted controller	2	**0.4982**	0.5426	0.4538
Controller without fragment merge	2	0.4967	0.5439	0.4494
Four-system character-probability combination	2	0.4825	0.5309	0.4342
GPT-5.5 + boundary control	1	0.4786	0.5175	0.4396
Direction-routed controller	1	0.4772	0.5200	0.4343
GPT-5.5 proposal	1	0.4691	0.5137	0.4244
Gemini 3.6 Flash + boundary control	1	0.4456	0.4890	0.4022
Local Gemma-4-31B-IT + MARBERTv2, character union	0	0.4446	0.5045	0.3847
Local Gemma-4-31B-IT + MARBERTv2, span union	0	0.4409	0.5048	0.3770
Gemma-4-12B-IT SFT + MARBERTv2	0	0.4357	0.5062	0.3651
MARBERTv2, three-seed ensemble	0	0.3817	0.4535	0.3099

Table 16: Official blind-test scores for our Subtask 3 systems. The submitted controller is bold; the underlined row is the strongest system without API models.

Missed error. [ENG_MAU, id 60306_9] Source: *"If not, they will see what they can do."* MT: وإلا **عاد** يشوفو شـ يديرو (English gloss: "Otherwise, they will see what they can do.") (gold: the bold word, sociolinguistics; predicted: no span). The error is a single discourse particle in an otherwise dialect-looking sentence, so no controller stage can recover it after every proposer misses it.

Most frequent category confusion. [ENG_EGY, id 1199_1] Source: *"Did you hear? You heard his frankness; he is straight and honest."* MT: سمعت؟ سمعت صراحته؛ هو **شريف** وصادق. (English gloss: "Did you hear? You heard his frankness; he is honorable and honest.") (gold: sociolinguistics; predicted: semantics). The proposer treats the bold word ("honorable" for "straight") as a lexical-choice error, whereas the gold annotation treats it as a register or variety error. This sociolinguistics-to-semantics cell contains 90 of the 368 matched category errors.

Boundary widening can reduce precision. [ENG_EGY, id 6181_1] Source: *"Or at least remove the worry from his heart."* MT: أو على الأقل **أزِل** القلق من قلبه. (English gloss: "Or at least remove the worry from his heart.") (gold and base prediction: the bold word, sociolinguistics). MARBERTv2 widens the exact base span to include the underlined word, adding six false-positive characters and reducing row-level character-mask F1 from 1.0 to 0.571. This is the example-level counterpart of widening's metric sensitivity in Table 17.

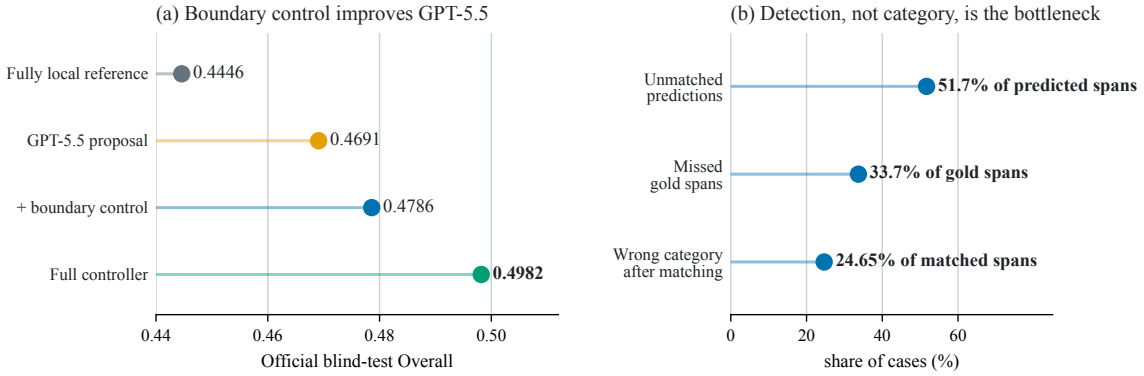


Figure 4: Subtask 3 controller analysis. (a) Official blind-test Overall for the fully local character-union hybrid and three proposer-controller stages. (b) Five-fold out-of-fold error rates for the pre-test direction-routed system: the local Gemma-4-31B-IT/MARBERTv2 hybrid handles Mauritanian inputs, and GPT-5.5 with boundary widening handles the other directions. Each rate uses the printed denominator. Unmatched predictions and missed gold spans remain more common than category errors among already matched spans.

Split	Stage	Deployed Overall	Δ	MPP-micro	Δ
CV	GPT-5.5 proposal	0.4565		0.4659	
CV	+ boundary widening	0.4655	+0.0090	0.4678	+0.0019
CV	+ fragment merge	0.4724	+0.0069	0.4773	+0.0095
CV	local reference (span union)	0.4573		0.4610	
dev	GPT-5.5 proposal	0.4236		0.4315	
dev	+ boundary widening	0.4296	+0.0059	0.4301	−0.0014
dev	+ agreement-gated additions	0.4459	+0.0163	0.4500	+0.0199
dev	+ fragment merge	0.4494	+0.0035	0.4514	+0.0014

Table 17: Controller stages under the deployed scorer and MPP-micro on released rows with gold (CV: 1,263 rows; dev: 138 rows). Deltas compare consecutive stages within each block. The local reference is the Gemma-4-31B-IT/MARBERTv2 span-union hybrid scored on pooled out-of-fold predictions, not part of the preceding CV chain.

	Predicted category							
Gold	Gra	Mor	Ort	Pra	Sem	Soc	Missed	Support
Graphetics	1	0	0	0	0	0	1	2
Morphosyntax	0	57	3	0	31	13	82	186
Orthography	0	16	32	0	21	5	43	117
Pragmatics	0	1	1	4	30	14	41	91
Semantics	0	28	19	17	226	43	213	546
Sociolinguistics	0	17	10	9	90	780	392	1298
Spurious pred.	12	137	96	23	370	1135		1773

Table 18: Category confusion for GPT-5.5 proposals with MARBERTv2 boundary widening on 1,263 cross-validation rows, using greedy one-to-one matching for spans with nonzero character overlap. Gra, Mor, Ort, Pra, Sem, and Soc abbreviate the six LQM categories.

F Cross-Task Comparison Record

ID	Configuration	Score	Δ (vs. row)	Phase
Unconstrained ST1: fixed local candidates				
U0	Strongest standalone candidate (E4B)	32.2336	n/a	S
U1	Four-candidate MBR without additional jurors	33.0925	+0.8589 (U0)	P
U2	+ E2B reference-only juror	33.1950	+0.1025 (U1)	S
U3	+ Six API reference-only jurors (submitted)	**33.4891**	+0.2940 (U2)	S
U4	All eleven systems made output-eligible	33.2998	-0.1893 (U3)	P
Unconstrained ST1: candidate replacement				
U5	Submitted MBR with the original NileChat candidate replaced by the full-data three-seed soup	33.5246	+0.0355 (U3)	P
Constrained ST1: submitted routing				
C0	Three-seed NileChat soup for all countries	29.5969	n/a	S
C1	Route LY/SD to train + public-test adapter (submitted)	**30.4233**	+0.8264 (C0)	S
Constrained ST1: supervision and weight averaging				
C2	Train + development	29.2323	n/a	P
C3	+ Duplicated rows at LY/SD budget	29.0971	-0.1352 (C2)	P
C4	+ Authentic public LY/SD rows	30.1179	+0.8856 (C2)	P
C5	+ Duplicated rows at full-public-test budget	28.8483	-0.3840 (C2)	P
C6	+ Authentic complete public test	30.4733	+1.2410 (C2)	P
C7	Equal-weight soup of three full-data models	30.9726	+0.4993 (C6)	P
ST2: Gemma-4-31B-IT without fine-tuning				
T0	Zero-shot without retrieval	24.6898	n/a	S
T1	Target-dialect retrieval, k=32, with copy guard	27.9496	+3.2597 (T0)	S
T2	+ Guided generation (submitted)	**28.4011**	+0.4516 (T1)	S
ST3: semantic proposals and span control				
E0	Raw GPT-5.5 proposals	0.4691	n/a	S
E1	+ MARBERTv2 boundary widening	0.4786	+0.0095 (E0)	S
E2	+ Gemini widening / gated additions / empty-row rescue	0.4967	+0.0181 (E1)	S
E3	+ Same-category fragment merging (submitted)	**0.4982**	+0.0015 (E2)	S

Table 19: Cross-task blind-test comparisons. S: official evaluation-window submission; P: unofficial post-evaluation experiment on the same blind test. Scores are macro spBLEU for ST1, spBLEU for ST2, and Overall for ST3. Each Δ names its reference row. C3 to C6 independently modify C2; all five use seed-42 adapters. C7 averages three full-data models, distinct from C0. ST2 retrieval includes copy handling; E2 combines controller extensions. Effects are not independently additive. Deltas use unrounded scores except for ST3's four-decimal records. Submitted final scores are bold.